Born Sub-Prime: Long-Term Impact of Early-Life Credit Access

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Abstract

This paper provides evidence of a credit access trap in the U.S., where declines in credit availability lock borrowers into a low-score, low-access equilibrium. Exploiting merger-induced branch closures as plausibly exogenous shocks to local credit supply, we find that sub-prime individuals exposed to early reductions in available credit experience persistently worse credit outcomes, with declines in both access to unsecured credit and credit scores. Credit supply shocks also limit geographic mobility, lowering the probability of relocating to higher-income areas. Individuals with limited prior credit histories at the time of exposure to these shocks suffer the strongest adverse effects. These results highlight the potential for targeted interventions to improve credit access for young, sub-prime borrowers.

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1 Introduction

Credit access is a crucial determinant of economic mobility, shaping people's ability to smooth consumption, search for jobs, pursue education, and invest in housing (Guiso and Sodini, 2013; Solis, 2017; Bos et al., 2018; Braxton et al., 2024). A growing body of evidence shows that credit access differs systematically by race, geography, and parental credit records (Ghent and Kudlyak, 2016; Hendren et al., 2025), and that initial conditions in the credit market persist throughout the lifecycle (Bach et al., 2023). Do persistent differences in credit outcomes reflect early screening of borrowers' types, or are they the result of a *credit trap*? If poor credit outcomes restrict future borrowing and record-building, people who enter the credit market on different terms could experience widening inequality over the lifecycle, even if they are otherwise identical. As a result, policies that subsidize early credit access could yield self-sustaining gains for sub-prime individuals.

This paper uses large administrative data on U.S. borrowing records to shed light on the existence of a credit trap. We find that negative shocks to borrowing opportunities at young ages lead to large and persistent declines in credit scores, unsecured debt, auto loans, and mortgages among subprime borrowers—individuals with initial credit scores below 600. We also show that early credit access is a critical determinant of upward geographic mobility: borrowers hit by negative shocks are less likely to move from lower- to higher-income ZIP codes. Consistent with a credit trap interpretation, negative shocks significantly harm only entrants without prior histories, for whom limited credit availability hinders score-building. By contrast, entrants with established histories (often through parents' credit cards) and older cohorts are largely unaffected.

Finding exogenous shocks to credit supply is challenging, especially outside the context of microcredit.[1] Our identification follows Nguyen (2019), using bank branch closings from mergers in 2004–2007 as credit supply shocks. We exploit the fact that large bank mergers often involve overlapping branches in close proximity, some of which are closed during post-merger restructuring due to geographic redundancy rather than local economic conditions. These closures generate quasi-experimental variation in local credit availability. Access to credit through physical bank branches is particularly important for young individuals with low scores and thin histories. Although online credit was available in the mid-2000s, spatial proximity to bank branches remained an important determinant of credit access for both households (Argyle et al., 2022) and firms (Degryse and Ongena, 2005; Brevoort et al.,

[1] For example, see Karlan and Zinman (2009), Karlan and Zinman (2010), Attanasio et al. (2015), Augsburg et al. (2015), Crépon et al. (2015), Breza and Kinnan (2021), and De Giorgi et al. (2023).

2010).[2] We define a ZIP code as exposed to merger-induced branch redundancy if, prior to a 2004–2007 merger between two banks each with at least $2 billion in assets, both the acquiring and acquired banks operated branches within its boundaries. Individuals residing in such ZIP codes and aged between 18 and 22 in 2004 are classified as treated. We then compare the outcomes of treated individuals with those of control individuals, defined as people who were between 18 and 22 years old in 2004 and resided in ZIP codes that (1) are within the same county as ZIP codes exposed to merger-induced branch redundancy, and (2) where the merging institutions did not *both* operate branches. We follow Callaway and Sant'Anna (2021) and use a staggered difference-in-differences (DiD) model to estimate the effect of branch closures on credit outcomes up to ten years after exposure.[3]

Identification relies on the assumption that, absent the 2004-2007 mergers, young individuals residing in ZIP codes exposed to merger-induced branch closures would have experienced similar trends in credit availability as young individuals residing in unexposed ZIP codes. This assumption is credible for three reasons. First, exposure to merger-induced branch closures is unlikely to be driven by ZIP-code-level shocks: across mergers, only 9% of the acquiring banks' branches were located in ZIP codes where the acquired bank also operated, suggesting that local conditions in those ZIP codes were unlikely to drive merger decisions. Second, older individuals' credit outcomes are unaffected by branch closures. This indicates that the observed effects on younger borrowers are unlikely to be driven by local economic conditions, which would have impacted both age groups. Finally, our empirical strategy accounts for time-invariant differences between treated and control ZIP codes, requiring any confounding factors to vary both across space and over time to bias our estimates.

Prior to the mergers, the number of branches in ZIP codes exposed to merger-induced branch closures evolved similarly to those in unexposed ZIP codes. Following the mergers, however, exposed ZIP codes experience a decline of about 0.5 branches, a reduction that persists over the subsequent decade. Merger-induced branch closures significantly reduce credit availability for treated individuals. In the year following a merger, credit card ownership drops by 2 percentage points, corresponding to a 5.5% decline relative to the 2004 sample average, which serves as the baseline level for expressing the short-term effects of the shocks as percentages throughout the analysis. Total credit card limits decrease by $250 on impact and by as much as $1,500 ten years after treatment (20.2% of the 2016 sample average,

[2] Evidence from the Survey of Consumer Finances indicates that as of 2007, 40.7% of U.S. credit customers cited branch location as the main reason for choosing their bank (Anenberg et al., 2018). (Argyle et al., 2022) show that proximity to financial institutions affects auto loan interest rates and amounts.

[3] To be able to estimate the long-term effects of branch closures, our analysis focuses on mergers filed between 2004 and 2007.

which serves as the baseline level for expressing long-term effects as percentages throughout the analysis) This reduction in credit availability also leads to lower credit scores: ten years after the event, treated individuals have credit scores that are 10–12 points lower and are over 4 percentage points (20.3%) less likely to be identified as prime borrowers. The decline in credit scores is not driven by an increase in borrowers' riskiness, as the treatment has no significant effect on the likelihood of 90+ day delinquencies or Chapter 7 bankruptcies.[4] Branch closures also reduce mortgages and auto loans. Ten years after treatment, the average mortgage size declines by up to $20,000, a 14.6% reduction relative to baseline. The average auto loan size also decreases by $500 (16.6%) on impact, though this effect dissipates over time. Notably, treated individuals file more inquiries for these lines of credit; e.g., auto loan inquiries increase up by 0.3 (52%) in the long run. This suggests that treated borrowers need to submit more applications to obtain credit, potentially reflecting lenders' reduced willingness to approve loans for these individuals. We also find that branch closures reduce geographic mobility. In the short run, treated individuals are 2 percentage points (5%) less likely to move outside their ZIP code. In the long run, they are 4 percentage points (13.1%) more likely to relocate to a low-income ZIP code and 6 percentage points (19.2%) less likely to relocate to a high-income ZIP code.

To shed light on the mechanism behind our results, we study how branch closures differentially affect individuals depending on the amount of information already present in their credit reports at the time of treatment. Specifically, we categorize individuals based on whether they have *inherited credit histories*, as indicated by credit records that predate their entry into the credit bureau files.[5] We find that the negative effects of branch closures on credit scores and borrowing are driven almost entirely by individuals entering the credit market without pre-existing histories. Similarly, we find that the negative effects of branch closures on credit outcomes are smaller and shorter-lived for individuals aged 23–30 than for individuals aged 18–22. These findings are difficult to reconcile with a pure screening model, in which credit histories simply reflect underlying borrower quality. Instead, they point to the existence of a *credit access trap*: for individuals who have not already been screened, reduced credit access leads to worse outcomes over time, which in turn hinders the accumulation of credit history and limits future access.

Evidence on the existence of a credit access trap has key policy implications. In particular,

[4] While the likelihood of Chapter 13 bankruptcies marginally increases following treatment, the magnitude of this effect is far too small to account for the observed worsening in credit outcomes.

[5] Following Bach et al. (2023), individuals with inherited credit histories are defined as those with credit records predating the year in which they turned 17—most likely because they were listed as co-users on their parents' accounts while still minors.

if individuals randomly subjected to reduced credit availability experience persistently worse outcomes in the credit market, policy interventions that subsidize credit supply for young or disadvantaged individuals may help them escape bad borrowing equilibria, potentially paying for themselves and reducing persistent inequalities over the lifecycle.

We contribute to a large literature on economic mobility (Solon, 1992; Chetty et al., 2014, 2016; Jantti and Jenkins, 2015) by highlighting the role of early credit access as a determinant of future opportunities. Prior work, such as Ghent and Kudlyak (2016) and Bach et al. (2023), documents the persistence of credit outcomes across generations and over the lifecycle. We extend this work by providing credible causal evidence that reductions in early credit availability have persistent effects on individual borrowing outcomes.

Our findings indicate that restrictions in early credit access not only limit short-term borrowing but also lead to persistent reductions in future credit scores, borrowing, and geographic mobility. There are many ways in which credit availability can shape people's economic trajectories. First, credit can play a key role in consumption smoothing, particularly when savings are limited (Zeldes, 1989; Jappelli, 1990; Deaton, 1991; Keys et al., 2017; Hundtofte et al., 2019). Second, limited access to borrowing constrains investment in human capital (Wiederspan, 2016; Solis, 2017) and housing (Topel and Rosen, 1988; Laufer and Paciorek, 2022). Third, access to credit supports entrepreneurial activity among young individuals with limited capital (Evans and Jovanovic, 1989; Banerjee and Newman, 1993), and facilitate geographic mobility (Bilal and Rossi-Hansberg, 2021; Giannone et al., 2023; Molloy et al., 2022). Finally, credit scores not only influence individuals' ability to borrow but also affect employment opportunities (Bos et al., 2018; Ballance et al., 2020; Corbae and Glover, 2025), and access to housing rentals and utility services.

Our results on the role of branch closures in disrupting relocation to high-opportunity areas are particularly related to Bos et al. (2018), who use a natural experiment in Sweden to show that negative credit information decreases employment and geographic mobility. While Bos et al. (2018) examine the effects of changes in the information available to potential lenders and employers, our paper focuses on effects of an exogenous reduction in credit supply. Second, whereas their study revolves around labor market outcomes, we analyze a broad set of credit market outcomes and show that the negative effects of branch closures are primarily borne by individuals who are exposed to these shocks when they are young and have limited credit histories. Studies exploiting variation in the visibility of credit defaults to employers find heterogeneous or negligible effects on employment prospects in the U.S. (Ballance et al., 2020; Dobbie et al., 2020), but large adverse effects in Sweden (Bos et al., 2018), where defaults are rarer and therefore potentially more informative signals for screening. Our results complement recent work on the role of credit scores in labor market outcomes by showing that early disruptions in credit access significantly constrain the relocation of individuals to higher-income areas.

The remainder of the paper is organized as follows. Section 2 describes the data and explains how we combine information on bank branches and mergers with credit report data to construct our main sample. Section 3 outlines our empirical strategy. Section 4 presents the main results. Section 5 examines the underlying mechanisms and provides a set of robustness checks. Section 6 concludes. Additional results are provided in the Appendix.

2 Data

2.1 Bank Branches and Mergers

We use publicly available data from the Federal Deposit Insurance Corporation (FDIC) to geolocate bank activity in each ZIP code. FDIC data include information on the universe of banking institutions active in the U.S. and their branches. Importantly, for each active bank branch, we observe the ZIP code in which it operates and the identifier of the banking institution to which it belongs as of June 30 of each year, the date on which data on credit records from the Experian bureau are pulled (see Subsection 2.2). The same banking institution identifier is included in the FDIC Register of Business Combinations, which reports data on merger agreements among banks in the US. From these data, we can identify the date of each merger, the banking institutions involved in it, and their role (acquiring or acquired). Since our credit report data span 2004–2016 (see Subsection 2.2) and our goal is to examine how early credit access shapes borrowing over the lifecycle, we focus on mergers between July 1, 2004, and June 30, 2007, which enable us to tie branch closures to credit market outcomes over a horizon of 9 to 12 years. Moreover, since our interest lies in quasi-random branch closures rather than closures driven by local economic conditions, we restrict attention to mergers between large banks; i.e., those in which both the acquiring and acquired banks held at least $2 billion in pre-merger assets.

In Table 2, we report descriptive statistics for the banks involved in the mergers we consider. Out of a total of 21 mergers, the average acquiring bank has almost 600 branches operating in 460 ZIP codes, while the average acquired bank has 217 branches operating in 166 ZIP codes. Overall, acquiring banks are larger than the institutions they acquire, as

Table 1: Mergers Among Large Banks (2004 - 2007)

Acquiring	Acquired	Date
Sky Bank	The Second National Bank of Warren	02/07/2004
The Provident Bank	First Savings Bank	14/07/2004
Sovereign Bank	Compass Bank for Savings	23/07/2004
JPMorgan Chase Bank	Bank One, N. A.	13/11/2004
Fifth Third Bank	First National Bank of Florida	01/01/2005
Wachovia Bank, N. A.	SouthTrust Bank	03/01/2005
First Niagara Bank	Hudson River Bank & Trust Company	14/01/2005
North Fork Bank	GreenPoint Bank	22/02/2005
Associated Bank, N. A.	First Federal Capital Bank	19/02/2005
Sovereign Bank	Waypoint Bank	11/02/2005
National City Bank	The Provident Bank	04/03/2005
SunTrust Bank	National Bank of Commerce	22/04/2005
Bank of the West	Commercial Federal Bank	03/12/2005
TD BankNorth, N. A.	Hudson United Bank	31/01/2006
The Huntington National Bank	Unizan Bank, N. A.	01/03/2006
Sovereign Bank	Independence Community Bank	09/09/2006
Washington Mutual Bank	Commercial Capital Bank, FSB	01/10/2006
MB Financial Bank, N. A.	Oak Brook Bank	02/11/2006
Regions Bank	AmSouth Bank	04/11/2006
New York Community Bank	Penn Federal Savings Bank	02/04/2007
Citizens Bank	Republic Bank	28/04/2007

The table lists mergers among large banks between July 1, 2004, and June 30, 2007, involving institutions with more than $2 billion in pre-merger assets. It includes the names of the acquiring and acquired banks, along with the merger filing date reported by the FDIC.

measured by average deposits among the two types of institutions. Importantly, the average acquiring bank operates only 9% of its branches in a ZIP code where the acquired bank also operates at least one. The share of deposits in overlapping branches is comparable. This suggests that overlapping branches are a small portion of the acquiring bank's network and are not financially more important than other branches, as measured by the value of their deposits. Figure 11 in the Appendix illustrates the geographic distribution of counties with overlapping acquiring and acquired branches operating in the same ZIP codes, highlighting that such closures are more concentrated in densely populated areas.

Table 2: Merging Institutions (2005-2007)

	Acquiring	Acquired
N Branches	598.75	217.33
	(658.22)	(349.30)
N Zip	460.60	166.43
	(488.07)	(256.07)
Deposits	52,359.31	13,749.68
	(75,159.85)	(28,253.01)
Share Overlapping Branches	0.09	0.25
	(0.08)	(0.17)
Share Overlapping Zip	0.08	0.24
	(0.07)	(0.17)
Share Overlapping Deposits	0.10	0.22
	(0.11)	(0.17)
Observations	21	21

The table presents descriptive statistics for the institutions involved in the mergers listed in Table 1, showing the average and standard deviation of the variables in the first column, separately for acquiring and acquired banks.

2.2 Data from Credit Reports

We use proprietary credit report data from Experian, covering approximately 2.2 million individuals representing a 1% random sample of the U.S. population with a valid credit score in 2010. The dataset spans the years 2004 to 2016, with reports drawn annually as of June 30. The data include basic demographic information such as date of birth and ZIP code of residence, along with about 400 credit-related variables. These capture a wide range of credit outcomes, such as the number of credit cards, auto loans, and mortgages; the balances on credit card, auto loan, and mortgage accounts; credit card limits; as well as indicators of delinquency, default, and bankruptcy.[6]

Since our goal is to examine how early lack of credit availability affects borrowing over the life cycle for individuals entering the credit market with low credit scores, we focus on the subsample of individuals who, in 2004, were 22 years old or younger and had a credit score below 600. These criteria are satisfied for $51,284$ individuals with complete histories.[7] Additionally, we restrict the sample to individuals who, at the time of a merger between large banks occurring between July 1, 2004, and June 30, 2007, lived in a *county* where both the acquiring and acquired banks had at least one branch in a common ZIP code. This leaves us with a final sample size of $22,890$ individuals.

Table 3 reports summary statistics for the subsample of individuals of interest in 2004, 2010, and 2016. For 2004, statistics are reported separately for individuals residing in ZIP codes exposed to mergers and those not exposed. Panel A presents demographic characteristics, credit scores, and borrowing activity, pooling across all types of credit lines.[8] We observe credit outcomes for a panel of 22,890 individuals over 12 years, covering their twenties and early thirties. By definition of our sample, these individuals start with low credit scores; in 2004, the average credit score is 526. As already noticed in Bach et al. (2023), initial credit scores are remarkably persistent: By 2016, only 21% of individuals in our sample exceed the threshold of 660, which Experian defines as a *good* credit score, and the average score remains below 600. Interestingly, between 2004 and 2016, while the number of credit lines and their balances increase by a factor of 3 and 10, respectively, the average number of delinquent credit lines remains stable around 0.1. Panel B focuses on credit cards. In 2004, 36% of the individuals in our sample have at least one credit card. By 2016, this share is increases to 54%. On average, credit card holders maintain multiple credit cards in any given, with 2.72 cards in 2004 and 4.31 in 2016. The average credit card holder had a credit limit of approximately $2,380 in 2004, increasing to $13,600 by 2016. Average utilization in 2004 is high, with cardholders using 85.3% of their available credit, declining to 52.9% in 2016. This pattern is mirrored by a reduction in the number of credit cards delinquent by more than 90 days, which falls from 0.56 in 2004 to 0.29 in 2016.

When comparing individuals in treated and untreated ZIP codes in 2004, the two groups appear similar in demographics, credit scores, and borrowing activity. However, treated individuals are more likely to reside in higher-income ZIP codes. While the similarity in

[6]For other papers using Experian's data on credit reports, see Albanesi and Vamossy (2019), De Giorgi et al. (2021), Bach et al. (2023), De Giorgi et al. (2024), and De Giorgi and Naguib (2024).

[7]We drop $3,177$ individuals not observed in all periods.

[8]Table 5 in the Appendix details the number of credit lines separately for mortgages and auto loans.

credit outcomes is reassuring, mergers tend to occur more frequently in higher-income areas, likely because large banks are more prevalent there, making such mergers more feasible. This pattern motivates our empirical strategy, which exploits time variation in exposure to mergers to avoid bias from time-invariant differences between the two groups, as discussed in Section 3.

Overall, our sample consists of individuals with low credit scores and limited access to credit at the start of the panel. Among those who obtain credit cards, we observe a substantial increase in credit limits over time, accompanied by a gradual decline in utilization rates. This is also reflected in a decrease in the number of delinquent cards, suggesting that these individuals improve their ability to manage credit lines over time.

3 Empirical Strategy

Our empirical strategy revolves around identifying the effect of negative shocks to early credit availability on borrowing over the lifecycle. We leverage merger-induced bank branch closures as an exogenous source of variation in local credit availability, following Nguyen (2019).

Physical bank branches facilitate credit access by lowering the search costs consumers face. Even in the era of online banking, empirical studies consistently find that proximity to a branch significantly influences access to credit (e.g., Degryse and Ongena (2005) and Argyle et al. (2022)). In the period when the mergers we study take place (2004–2007), and especially for young individuals with low credit scores and short credit histories, proximity to bank branches is likely to be particularly important.Our results confirm this interpretation.

To identify supply-side shocks in credit, we use merger-driven branch closures as an exogenous reduction in local credit availability. To build intuition, consider a merger between two large banks. Before the merger, some ZIP codes may host branches from both institutions. Afterward, overlapping branches are closed not because of shifts in local credit demand, but to reduce costs arising from geographic redundancy under common ownership. To exploit these events as natural experiments, we define a ZIP code as exposed to merger-induced branch redundancy if, prior to a 2004-2007 merger between two banks with at least $2 billion in assets, both the acquiring and the acquired banks operated branches within its boundaries. We define a ZIP code as unexposed to merger-induced branch redundancy if it satisfies two conditions: (1) it is located within the same county as ZIP codes exposed to merger-induced branch redundancy and (2) the merging institutions did not *both* operate branches within its boundaries. We classify an individual as treated if he or she resided in

Table 3: Summary Statistics of Credit Outcomes (2004–2016)

	2004		2010	2016
Exposed to Merger	*NO*	*YES*	*Both*	*Both*
A: Customer Information				
Age	20.64	20.66	26.65	32.65
	(1.22)	(1.22)	(1.22)	(1.22)
Credit Score	526.02	526.88	553.47	583.53
	(48.27)	(46.99)	(80.64)	(92.10)
Good Credit Score (> 660)	0.00	0.00	0.12	0.21
	(0.00)	(0.00)	(0.32)	(0.41)
ZIP Income - Percentile	41.80	51.74	45.21	47.40
	(31.33)	(30.31)	(32.37)	(32.21)
N Open Credit Lines	1.53	1.45	2.27	4.20
	(2.36)	(2.21)	(3.13)	(5.06)
Balance	4,839.24	4,522.19	19,597.01	43,925.22
	(18,611.96)	(14,589.39)	(52,033.17)	(88,766.52)
N Credit Lines 30d+ Delinquent	0.10	0.10	0.05	0.07
	(0.37)	(0.36)	(0.28)	(0.35)
N Inquiries	3.42	3.00	2.84	2.91
	(2.73)	(2.43)	(2.40)	(2.59)
B: Credit Cards				
P Credit Card Holder	0.36	0.35	0.34	0.54
	(0.48)	(0.48)	(0.47)	(0.50)
N Credit Cards	0.98	0.93	1.17	2.33
	(1.86)	(1.78)	(2.14)	(3.47)
Credit Limit, CC	862.70	828.62	2,222.70	7,357.94
	(3,054.40)	(3,006.40)	(6,221.81)	(15,378.57)
Utilization %, CC	85.30	85.28	56.76	52.99
	(50.60)	(51.18)	(42.87)	(37.89)
P Credit Card 90d+ Delinquent	0.30	0.31	0.19	0.15
	(0.46)	(0.46)	(0.39)	(0.36)
N Inquiries, CC	1.35	1.21	1.05	1.23
	(1.53)	(1.36)	(1.32)	(1.58)
Observations	18953	3937	22890	22890

The table reports summary statistics for individuals aged 18–22 with low credit scores in 2004, showing the mean and standard deviation of each variable listed in the first column. Panel A summarizes demographic characteristics, credit scores, and overall borrowing activity, pooling across all credit types. Panel B focuses specifically on credit cards. All averages are computed over the full sample, except for credit card utilization, which is defined only for years when an individual holds a credit card. For 2004, statistics are reported separately for individuals exposed and not exposed to a merger between July 1, 2004, and June 30, 2007. For 2010 and 2016, statistics are computed for the full sample.

an exposed ZIP code between the ages of 18 and 22, and as control (or untreated) if they resided in an unexposed ZIP code during the same age range., controlling for the number of active branches prior to the event.

To further address potential endogeneity concerns, we compare outcomes between exposed and unexposed areas, and between treated and control individuals, using a difference-in-differences design. The validity of this approach rests on the identifying assumption that, absent the merger, exposed and unexposed ZIP codes would have followed similar trends in credit availability.

This assumption is credible for two main reasons. First, on average across mergers, only 9% of an acquiring bank's branches are located in ZIP codes where the acquired bank also operates a branch, suggesting that local conditions in those ZIP codes do not influence merger decisions. Second, the difference-in-differences design accounts for time-invariant differences between exposed and unexposed ZIP codes. Thus, for our estimates to be biased, confounding factors would have to vary not only across locations but also over time. The absence of pre-trends in bank activity supports the validity of our identification strategy (Figures 2 and 3).

As an illustration of our empirical strategy, Figure 1 shows the location of bank branches in Los Angeles County in 2005 for Washington Mutual Bank, Commercial Capital FSB, and other large banks. After the 2006 merger between Washington Mutual and Commercial Capital FSB, ZIP codes hosting branches from both the acquiring and the acquired bank are classified as treated (darker shaded areas), while ZIP codes with branches from large banks but without overlap are classified as part of the control group (lighter shaded areas).

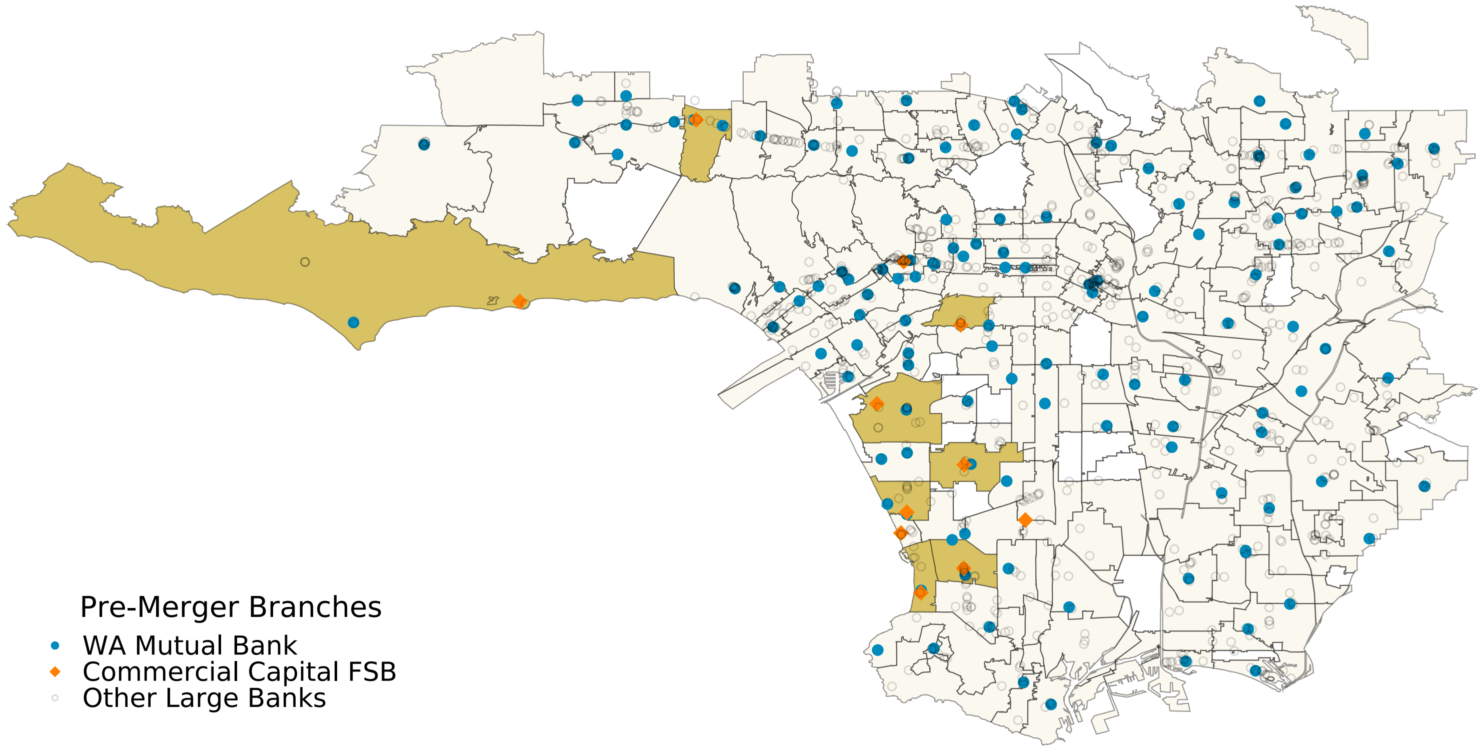


Figure 1: Bank branches in Los Angeles County, CA 2005.

The map shows the locations of bank branches in the ZIP codes of Los Angeles County, CA, in 2005. Blue dots represent branches of the acquiring bank, Washington Mutual Bank, while orange dots denote branches of the acquired bank, Commercial Capital FSB. Shaded black dots indicate branches of other large banks. Dark-shaded areas represent ZIP codes with overlapping branches from both WA Mutual Bank and Commercial Capital FSB, which constitute the treated group. Light-shaded areas represent other ZIP codes with large banks, which are included in the control group. Source: FDIC, authors' own calculations.

To compare the evolution of outcomes between exposed and unexposed ZIP codes, we follow Callaway and Sant'Anna (2021) and estimate the following staggered DiD model:

$$B_{r,t} = \alpha_r^\tau + \gamma_t^\tau + \sum_l \beta_l^\tau \left(Z_r^\tau \cdot \mathbf{1}\left\{ t = \tau + l \right\} \right) + \lambda_t^\tau \mathbf{X}_r + \epsilon_{r,t}^\tau. \tag{1}$$

Here, $B_{r,t}$ represents the outcome of interest for ZIP code r in year t, α_r^τ and γ_t^τ are ZIP code and year fixed effects, respectively, and $\mathbf{X}_r$ is a vector of ZIP code characteristics in the baseline year 2004. In our preferred specification, $\mathbf{X}_r$ includes the number of large banks operating a branch in ZIP code r in 2004.[9] The variable Z_r^τ equals 1 for exposed ZIP codes and 0 for unexposed ZIP codes, where $\tau \in \{2004, 2005, 2006, 2007\}$ denotes the year of the merger. The coefficients β_l^τ capture the average treatment effect for exposed ZIP codes in year $\tau + l$. Following Callaway and Sant'Anna (2021), we estimate the model separately for each merger year τ, including only never-exposed ZIP codes in the comparison group, and present the results in an event-study format by averaging the β_l^τ across merger years τ for each lead or lag l relative to the merger date.

To compare the evolution of outcomes between treated and untreated individuals, we estimate the following staggered DiD model:

$$Y_{i,t} = \mu_i^\tau + \kappa_t^\tau + \sum_l \delta_l^\tau \left(Z_{r(i,\tau-1)} \cdot \mathbf{1}\left\{ t = \tau + l \right\} \right) + \phi_t^\tau \mathbf{X}_{r(i,\tau-1)} + e_{i,t}^\tau. \tag{2}$$

Here, $Y_{i,t}$ denotes the outcome of interest for individual i in year t, while μ_i^τ and κ_t^τ represent individual and year fixed effects, respectively. The function $r(i, \tau - 1)$ indicates the ZIP code where individual i resided in year $\tau - 1$. The variable $Z_{r(i,\tau-1)}$ equals 1 for individuals living in exposed ZIP codes in the year prior to a merger, and 0 for individuals who never lived in an exposed ZIP code in any merger year. $\mathbf{X}_{r(i,\tau-1)}$ denotes the same vector of ZIP code characteristics included in (1), measured for the ZIP code where individual i resides in the year prior to the merger. As before, the coefficients δ_l^τ capture the average treatment effect for individuals living in exposed ZIP codes at time $\tau + l$. Following Callaway and Sant'Anna (2021), we aggregate results across different merger years by averaging the δ_l^τ.

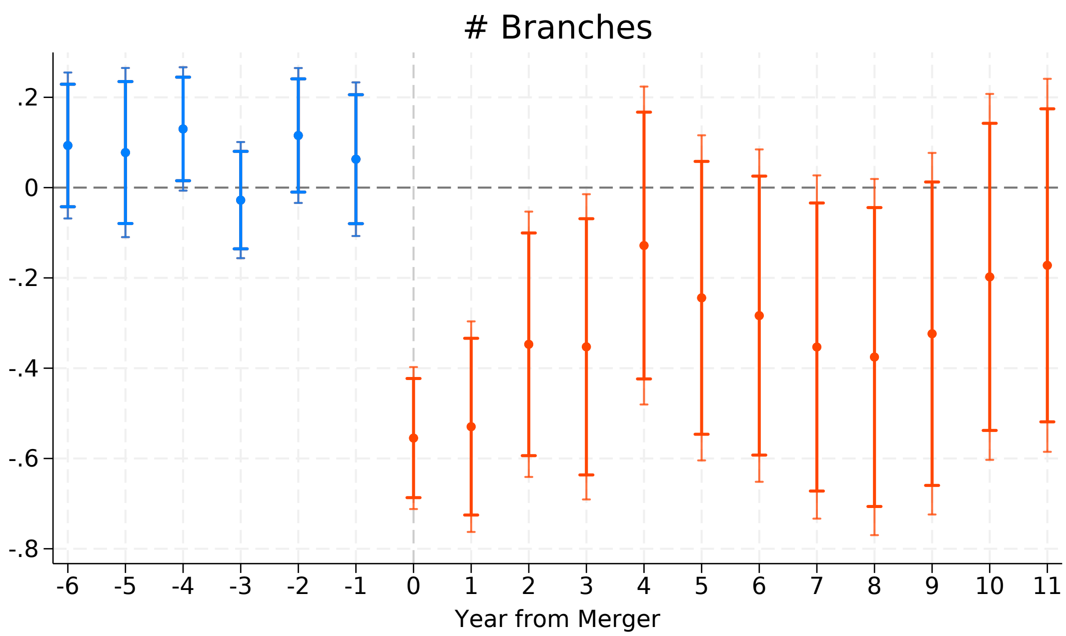
[9] The results are robust to the inclusion of additional controls based on 2000 Census ZIP code demographics, including population, median income, and the share of households below the poverty line.

4 Results

4.1 Mergers and Branch Consolidations

Do mergers between large banks result in branch closures that disproportionately affect ZIP codes where both the acquiring and the acquired bank had branches prior to the merger? To answer this question, we estimate model (1) using the number of branches active in a ZIP code in a given year as the dependent variable. Figure 2 presents the estimated average treatment effects in an event-study format, comparing trends in branch activity between exposed and unexposed ZIP codes before and after the merger.

Figure 2: Number of Open Branches



The figure shows the estimated impact of merger taking place at year $t = 0$ on the number of branches in exposed ZIP codes. Estimates are based on model 1 and aggregated ATTs from mergers between July 2004 and June 2007, following Callaway and Sant'Anna (2021). Bars display 95% and 90% confidence intervals.

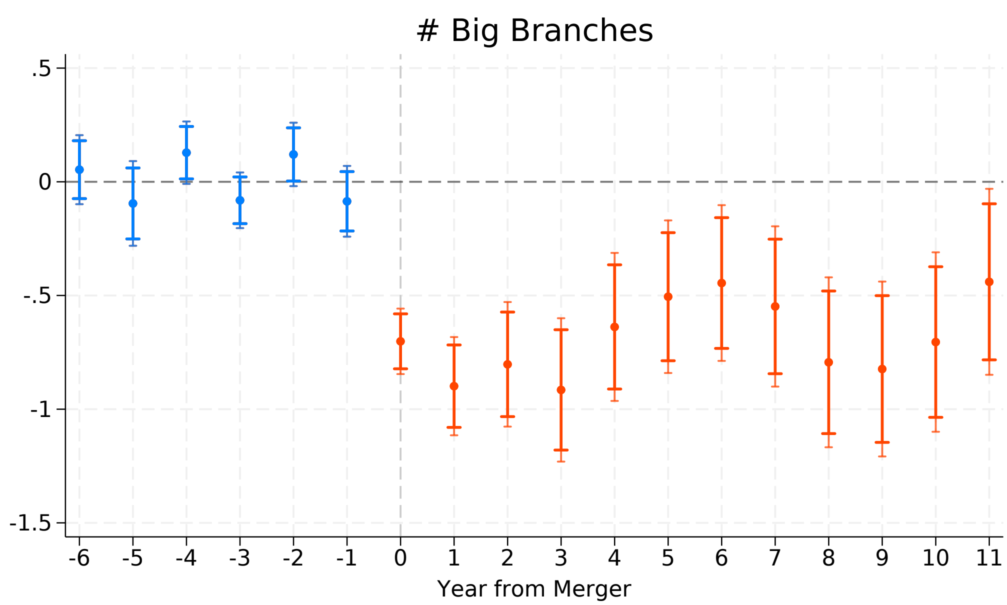
The number of branches in exposed ZIP codes declines significantly following mergers. We estimate that areas served by both the acquiring and the acquired banks experience a reduction of approximately 0.5 branches over the three years following these mergers.[10] This

[10]This estimate is consistent with Nguyen (2019), who find a reduction of roughly 0.3 branches on average in the first two years following mergers. Note that our sample covers mergers occurring in different years

suggests that mergers between large banks lead to branch closures in areas where both banks have a presence. Such closures can reduce credit access due to increased average distance to a branch, greater crowding at local branches, or decreased competition among credit providers. Exposed and unexposed areas exhibit similar trends before mergers take place, reinforcing the credibility of the parallel trends assumption.

Figure 3 replicates the same design, using the number of branches from large banks only (those with $2 billion or more in pre-merger assets) as the dependent variable. The short-run effect of mergers is qualitatively similar: following a merger, areas with overlapping branches are more likely to experience branch closures, leading to a reduction of about 0.8 branches. However, this effect is larger and persists in the long run, up to 11 years after mergers take place. This pattern suggests that while mergers between large banks initially result in branch closures, over time smaller institutions enter these areas at a higher rate than in unexposed ZIP codes, partially offsetting the overall decline in total branches.

Figure 3: Number of Open Branches, Large Banks Only



The figure shows the estimated impact of merger taking place at year $t = 0$ on the number of branches from large banks ($\geq$ $2 billion in pre-merger assets) in exposed ZIP codes. Estimates are based on model 1 and aggregated ATTs from mergers between July 2004 and June 2007, following Callaway and Sant'Anna (2021). Bars display 95% and 90% confidence intervals.

Overall, the results support the notion that mergers among large banks are unlikely

and involving different banks than those studied in Nguyen (2019), and we also use a different cutoff for bank size in the estimation.

to be driven by local economic conditions. First, before mergers take place, bank activity followed parallel trends in exposed and unexposed ZIP codes. If mergers were responding to differential pre-existing demand, we would expect to see changes in branch activity before the mergers, as banks can relocate branches at any time. Second, while the number of large bank branches is persistently reduced after the mergers, smaller financial institutions increase their presence disproportionately in exposed ZIP codes in subsequent years, partially mitigating the reduction in credit access. This pattern makes it unlikely that the mergers and subsequent branch closures were driven by deteriorating local demand, which would have reduced, rather than increased, the presence of smaller institutions.

4.2 Unsecured Credit and Credit Scores

We now turn to the main question of this paper: how do negative shocks to early credit access affect the credit outcomes of sub-prime individuals? To address this, we estimate the model in equation (2), using various credit outcomes as the dependent variable. We interpret these results as reduced-form estimates, leveraging merger-induced branch closures as an exogenous shock to local credit supply. Consequently, the effects presented in this section can be interpreted as the impact of an average reduction of approximately 0.5 branches available to young, sub-prime individuals.

Figure 4 presents the main results on access to credit cards. The top-left panel shows the treatment effect on the probability of holding a credit card. We find that merger-induced reductions in bank branch availability decrease the likelihood of holding a credit card compared to individuals in unexposed ZIP codes.[11] In absolute terms, the impact is 2 percentage points on impact, equivalent to 5.6% of the sample average in 2004, and persists for up to 10 years, although its statistical significance diminishes over time.

The top-right panel documents an average reduction of approximately 0.2 of a credit card line (20% relative to 2004 average) relative to unexposed individuals, an effect that remains statistically significant throughout the 11-year post-merger period. The bottom-left panel shows a gradual decline in the total amount of credit available to exposed individuals: available credit decreases by around \$500 over the three years following the mergers and continues to decline, reaching an overall reduction of approximately \$1,500.

Turning to default behavior, the bottom-right panel shows mixed evidence on 90-day credit card delinquencies, with a modest medium-term decrease but no significant short- or

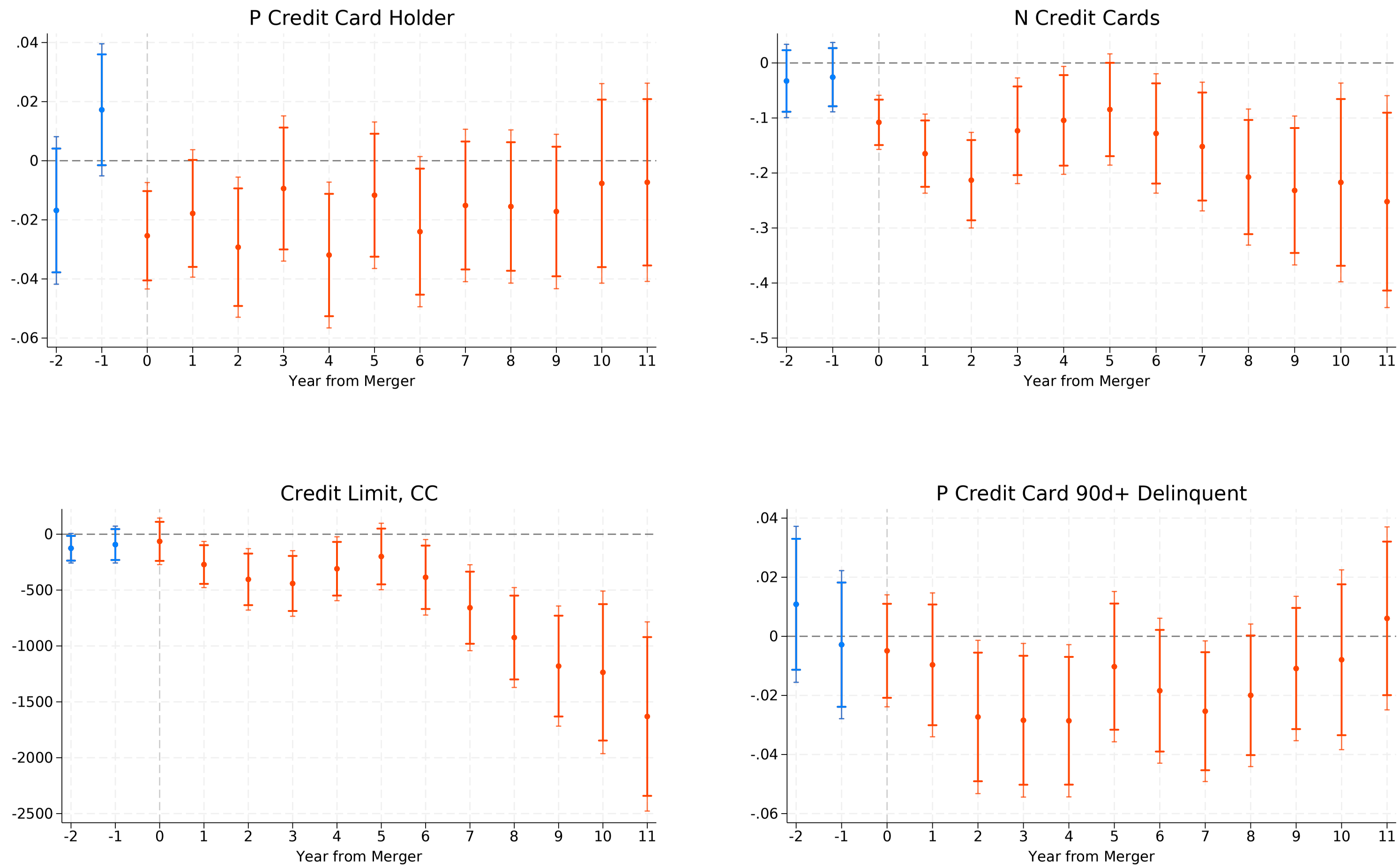
[11] To reduce noise potentially arising from secured credit cards with low limits, we define credit card holders as individuals with a credit limit exceeding \$500. Results remain qualitatively unchanged when including individuals with lower credit limits, as shown in Appendix Figure 13.

long-term impact. This pattern is consistent with results for bankruptcy filings in Figure 12 in the Appendix, where we find no significant effect on Chapter 7 bankruptcies and only a small, non-monotonic effect on Chapter 13 filings.

Overall, these results suggest that branch closings significantly affect the amount of unsecured credit that young, sub-prime individuals can access through credit cards. In addition to providing a tool for smoothing income and expense shocks, credit cards are one of the primary tools for building a credit history through consistent usage and timely payments. Hence, reduced access may hinder credit score development and, in turn, limit access to other forms of credit in the future. We investigate these outcomes next.

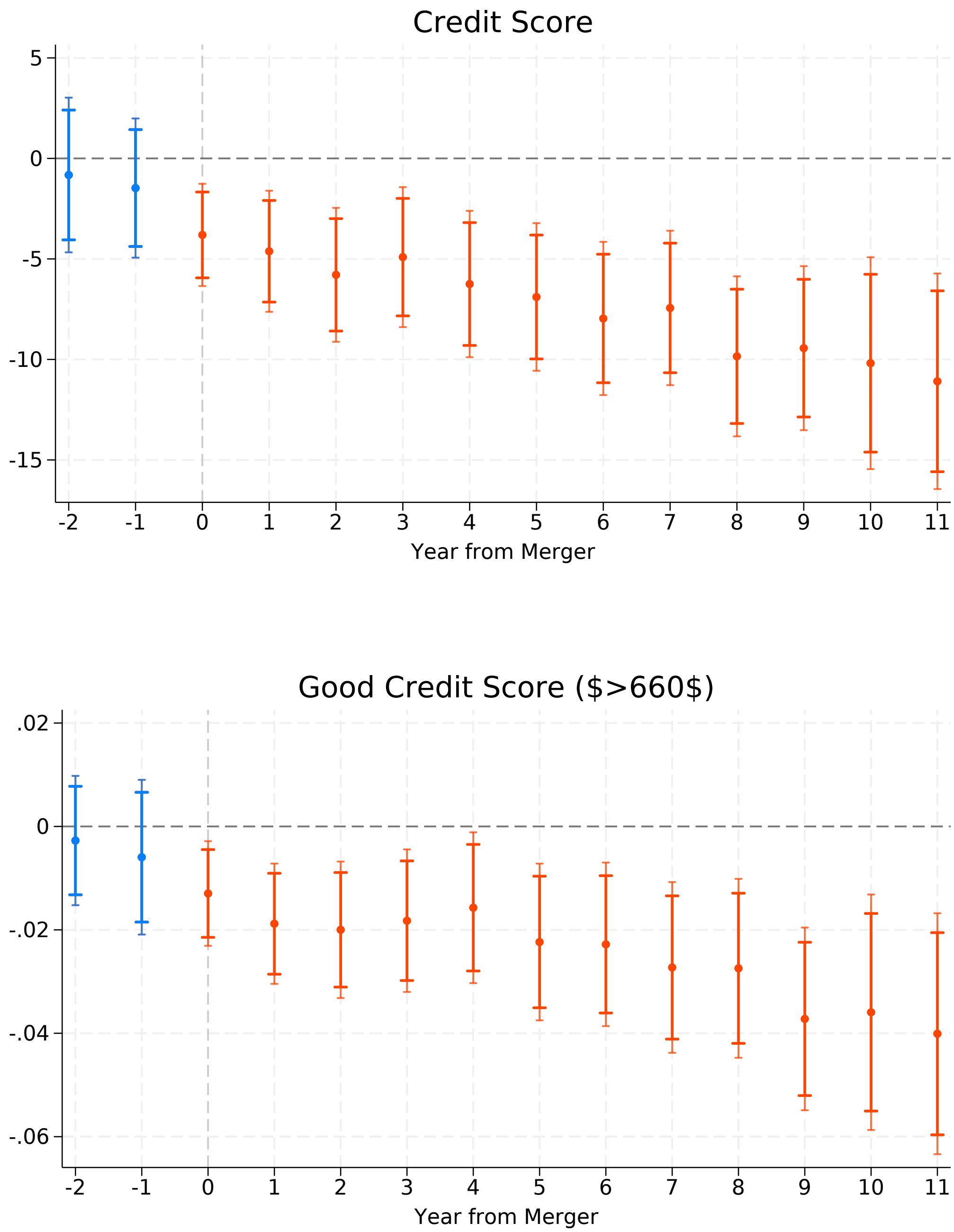
The two panels of Figure 5 illustrate the dynamic effects of the negative credit supply shock on credit scores. On average, young individuals experiencing a reduction in credit availability see their credit scores decline by 3–6 points immediately after the merger, with the effect growing over time to approximately 12 points 11 years post-merger. This effect is not only statistically significant but also economically meaningful, as it can shift individuals from one credit profile category to another. Specifically, the bottom panel of Figure 5 shows the change in the probability of having a credit score above 660, the threshold Experian defines as *Good*. This probability drops by 1 percentage point on impact and continues to decline to 4 percentage points after 10 years, representing a reduction of roughly 20% of the 2016 sample average (Table 3). These results highlight that early reductions in credit availability have persistent consequences for the creditworthiness of young individuals, with potential implications for their future access to credit and broader financial opportunities.

Figure 4: The Effect of Credit Shock on Credit Cards



Each plot shows the estimated impact of a merger taking place at year $t = 0$ on different measures of credit card usage for individuals living in exposed ZIP codes. Estimates are based on model 2 and aggregated ATTs from mergers between July 2004 and June 2007, following Callaway and Sant'Anna (2021). Bars display 95% and 90% confidence intervals. The dependent variable is reported above each plot.

Figure 5: The Effect of Credit Shock on Credit Score



Each plot shows the estimated impact of a merger taking place at year $t = 0$ on credit score for individuals living in exposed ZIP codes. Estimates are based on model 2 and aggregated ATTs from mergers between July 2004 and June 2007, following Callaway and Sant'Anna (2021). Bars display 95% and 90% confidence intervals. The dependent variable is reported above each plot.

4.3 Mortgages and Auto Loans

Here, we study the effect of merger-induced branch closures on mortgages and auto loans. The left panels of Figure 6 depict the impact of merger-induced branch closures in credit availability on mortgages. While we do not observe a statistically significant relative decrease in the probability of holding a mortgage, or change in the likelihood of applying for one (as measured by hard inquiries recorded by Experian), we do observe that, among individuals who do obtain a mortgage, the original loan amount declines. Specifically, mortgage holders experience a reduction of approximately \$5,000 in the first two years following the merger. This effect becomes noisier and loses statistical significance in the intermediate years, but re-emerges in the final year of the sample: treated individuals hold mortgage loans that are roughly \$20,000 lower than those of individuals in unexposed ZIP codes, corresponding to 14.6% of the average mortgage amount in 2016.

The right panels of Figure 6 show the corresponding estimates for auto loans. Individuals exposed to negative shocks to credit availability experience a reduction in the probability of holding an auto loan between the second and eighth year after the merger, equivalent to roughly 10% of the 2004 sample average. However, this effect loses statistical significance in subsequent years, and the point estimate gradually converges to zero. In contrast, exposed individuals exhibit a substantial and persistent increase in the number of auto loan inquiries. Among those who obtain an auto loan, the average original loan amount decreases by approximately \$500 (9.4% of the 2010 average) during the first eight years post-merger, though this difference diminishes over time.

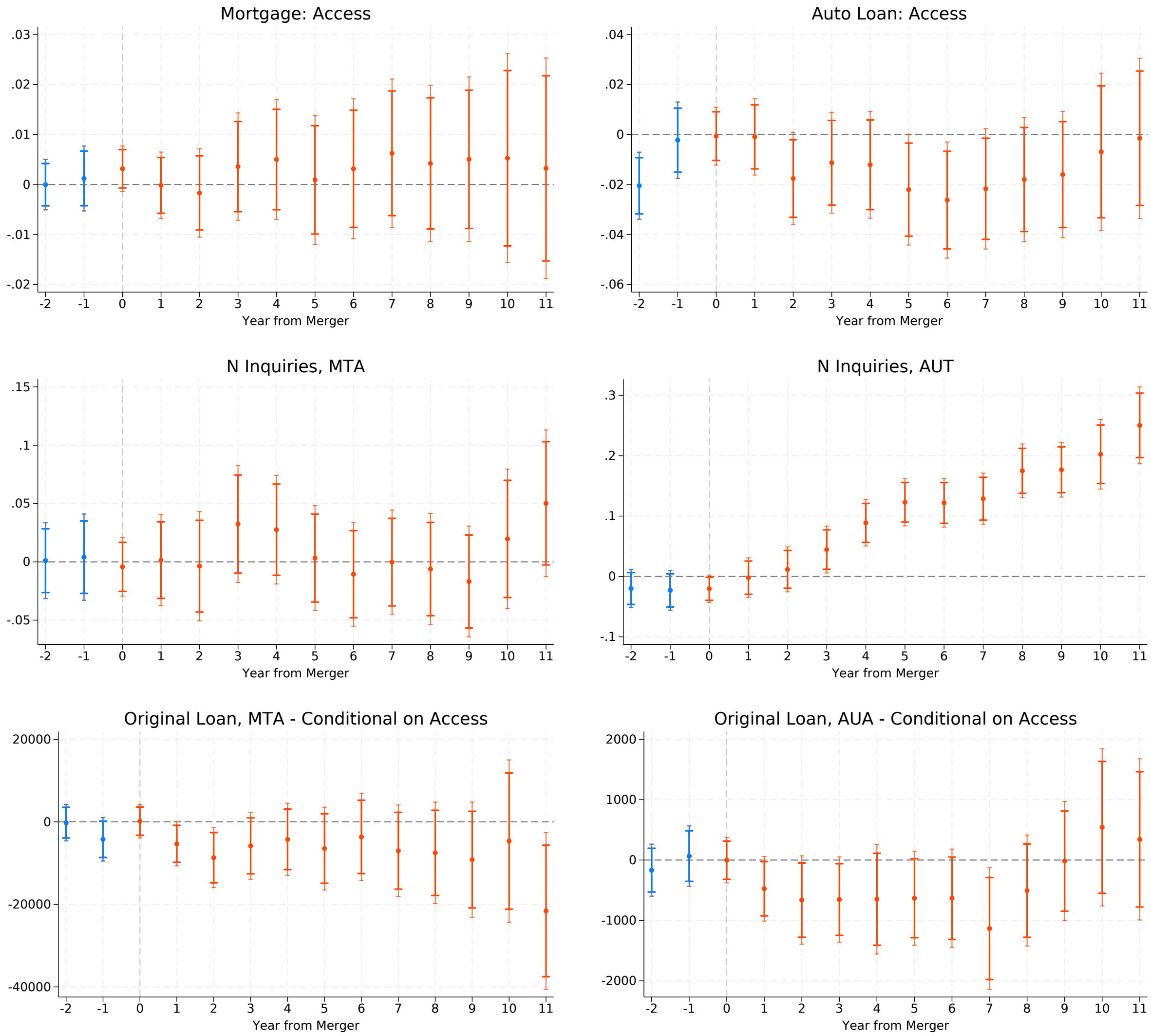


Figure 6: The Effect of Credit Shock on Mortgages and Auto Loans

Each plot shows the estimated impact of a merger taking place at year $t = 0$ on different measures of access to mortgages and auto loans for individuals living in exposed ZIP codes. Estimates are based on model 2 and aggregated ATTs from mergers between July 2004 and June 2007, following Callaway and Sant'Anna (2021). Bars display 95% and 90% confidence intervals. The dependent variable is reported above each plot. Mortgage original loan amounts are reported conditional on having a mortgage at any time between 2004 and 2016. Auto original loan amounts are reported unconditionally on having a loan.

4.4 Geographic Mobility

The ability to move to a new location is a crucial aspect of economic mobility, as it allows individuals to access better job opportunities and improve their overall economic conditions. In this section, we examine how merger-induced branch closures affect the likelihood of moving to a different address and the average income level at the new location.

Before turning to the results, we present descriptive statistics on the geographic distribution of individuals in our sample over time. Using publicly available IRS data, we compute the average income level of each ZIP code and rank them based on their position in the national distribution for each year. Table 4 shows that our sample is concentrated in lower-income areas: 42% of the individuals live in ZIP codes in the bottom income tercile in 2004, and 39% do so in 2016.[12] As shown in panel A of Figure 7, individuals in our sample tend to relocate relatively early. By 2005, 40% had already moved out of their 2004 ZIP code, and by 2010, the cumulative share of movers had nearly reached its peak at 80%.

Since access to credit can influence migration decisions through multiple channels, we now examine the effects of negative credit supply shocks on geographic mobility. The top panel of Figure 8 shows an average drop of 2 percentage points (5% of the sample average in 2005) in the probability of moving to a different ZIP code in the first four years following exposure to a credit supply shock. Moreover, beyond the average response, there are offsetting effects on mobility to higher-income versus lower-income ZIP codes, as measured by IRS average income. The bottom panels of Figure 8 show that in the long run, individuals exposed to credit supply shocks are 4 percentage points (13%) more likely to have moved to a low-income ZIP code and 6 percentage points (19%) less likely to have moved to a high-income ZIP code. Overall, exposed individuals delay relocating away from their original ZIP code in the first years after the shock, and, once they move, tend to relocate to lower-income areas. These results are consistent with our prior findings on mortgage amounts, since high-income ZIP codes are more likely to have higher house prices. This supports the idea that limited access to credit can constrain individuals' ability to move to locations offering better economic opportunities, as they may be unable to finance housing or other relocation-related expenses. We should note that this analysis is conditional on the endogenous decision to move. Therefore, the results should be interpreted as suggestive. In fact, noticeable pre-trends suggest that our estimates may be downward biased.

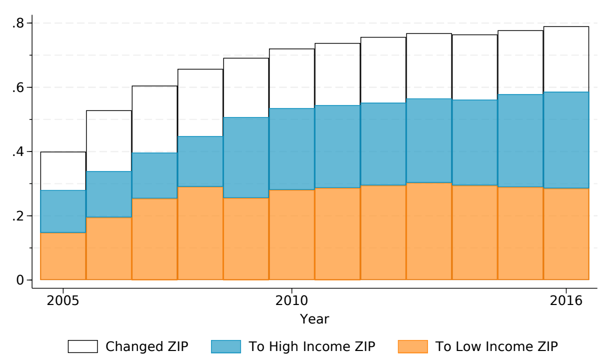
[12]Moreover, as shown in Appendix Figure 15, this initial concentration is persistent: by 2016, two out of three individuals who lived in a low-income ZIP code in 2004 are in a similarly ranked ZIP code in 2016.

Table 4: Geographic Location and Mobility (2004–2016)

	2004	2010	2016
ZIP Income - Percentile	43.53	45.21	47.40
	(31.38)	(32.37)	(32.21)
Bottom 33% ZIP - Income	0.42	0.43	0.39
	(0.49)	(0.49)	(0.49)
Top 33% ZIP - Income	0.30	0.34	0.36
	(0.46)	(0.47)	(0.48)
Changed ZIP	0.00	0.72	0.79
	(0.00)	(0.45)	(0.41)
Changed ZIP, to Low Income ZIP	0.00	0.28	0.29
	(0.00)	(0.45)	(0.45)
Changed ZIP, to High Income ZIP	0.00	0.25	0.30
	(0.00)	(0.44)	(0.46)
Observations	22,890	22,890	22,890

The table reports summary statistics for ZIP code characteristics in 2004, 2010, and 2016. It shows sample means and standard deviations (in parentheses) for the variables listed in the first column. ZIP codes are classified into percentiles of the year-specific income distribution, based on IRS data. Changes are measured relative to 2004. High-income ZIP codes correspond to the top tercile of the income distribution, and low-income ZIP codes to the bottom tercile.

Figure 7: Mobility Rates



The figure shows the probability of living in a ZIP code different from that in 2004 (empty bars), as well as the probability of living in a different ZIP code in the bottom (orange) or top (blue) income tercile. ZIP codes are classified into percentiles of the year-specific income distribution, based on IRS data.

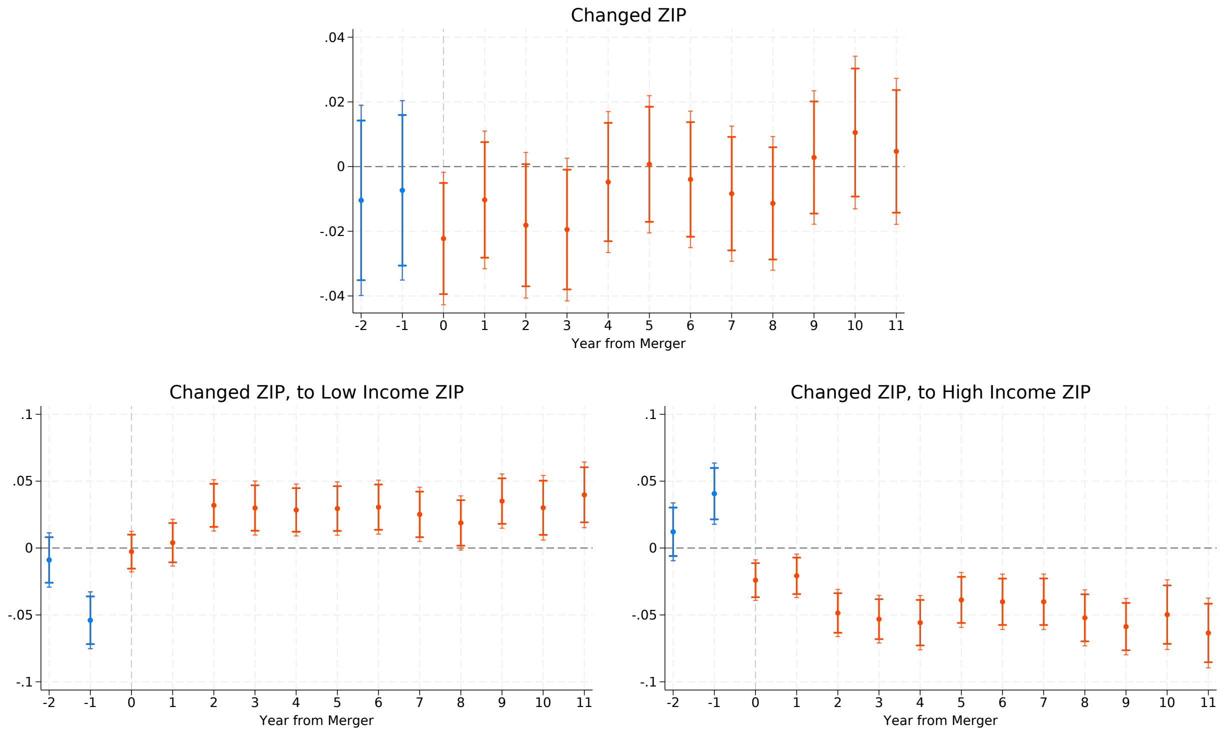


Figure 8: Migration and the Effect of Credit Shocks on Geographic Mobility

Each plot shows the estimated impact of a merger taking place at year $t = 0$ on different measures of geographical mobility for individuals living in exposed ZIP codes. Estimates are based on model 2 and aggregated ATTs from mergers between July 2004 and June 2007, following Callaway and Sant'Anna (2021). Bars display 95% and 90% confidence intervals. The dependent variable is reported above each plot.

5 Mechanism: The Role of Histories

In this section, we explore the heterogeneity of the effect of credit supply shocks by age and length of credit history, in order to validate the assumption behind our identification strategy and document an important mechanism driving our result: credit scores of individuals with shorter credit histories are more disrupted by early reductions in credit availability.

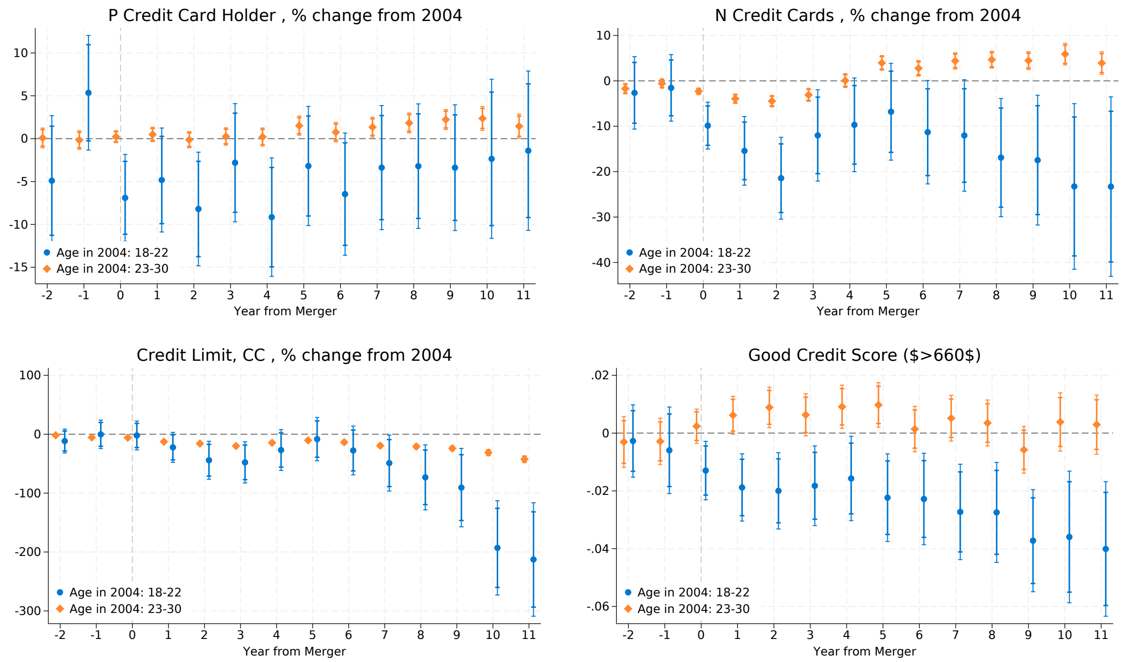
5.1 Older Versus Younger Cohorts

A potential concern for the validity of our research design involves the localized nature of our treatment. Specifically, a reduction in credit availability at the ZIP code level, even if uncorrelated with local economic conditions or demand factors, could still lead to a slowdown in the local economy. For instance, reduced access to credit for local businesses may dampen economic activity, which could, in turn, deteriorate the area's economic prospects. If so, our findings on the long-term consequences of credit shocks might reflect a decline in local economic opportunities, rather than the disruption of an individual's ability to build a history of timely payments.

To address this concern, we contrast the effect of merger-induced branch closures on our main sample of young individuals (ages 18–22 in 2004) with their effect on older subprime neighbors (ages 23–30 in 2004). The results are presented in Figure 9, where we report estimates of δ_l^τ from the estimation of model (2) separately for each age group. All panels show a much larger negative impact of branch closures on the younger cohort. The top-right and bottom-left panels in particular, show a statistically and economically significant difference, with the older cohort showing small or even positive effect on the number of credit cards and credit card limits, while the younger cohort shows strong negative impact. In terms of credit scores, the bottom right panel shows no statistically significant impact of branch closures on the probability of having a *Good* (above 660) credit score, compared to a significant drop for the young cohort.

This finding provides evidence against the possibility that merger-induced branch closures trigger a local economic downturn, as such a downturn would likely affect older age groups as well. Moreover, it indicates that the negative consequences of early credit shocks are concentrated among younger individuals, whose credit histories are shorter and less established. This pattern supports a 'credit access trap' interpretation, in which limited credit access locks individuals with short credit histories into low creditworthiness. We address this point directly in the next subsection by comparing the effect of credit supply shocks on individuals with and without an 'inherited' credit history from being co-users on family accounts.

Figure 9: The Effect of Credit Shock by Age



Each plot shows the estimated impact of a merger taking place at year $t = 0$ on different credit outcomes for individuals living in exposed ZIP codes. The effects are estimated separately for individuals aged 18-22 in 2004 and individuals aged 23-30 in 2004. Estimates are based on model 2 and aggregated ATTs from mergers between July 2004 and June 2007, following Callaway and Sant'Anna (2021). Bars display 95% and 90% confidence intervals. The dependent variable is reported above each plot.

5.2 Impact of Having Pre-Existing Credit History

Our main hypothesis is that exogenous reductions in early credit availability have long-term implications for sub-prime borrowers, as they prevent these individuals from building a history of timely payments. A key implication of this hypothesis is that the effect of credit supply shocks should be larger for individuals with a shorter existing credit history, for whom an additional credit line is likely more consequential. We test this implication by comparing the effect of merger-induced branch closures on customers with and without an inherited credit history.

We define individuals as having an inherited credit history if they have any recorded activity predating their 17^{th} birthday, based on the number of months since the opening of their oldest credit line. These longer credit histories are most likely a result of being assigned as a co-user on a family member's accounts while still a minor.[13]

[13]It is common for young individuals to open their first line of credit as an authorized user linked to the

The results, presented in Figure 10, indicate that the effect of merger-induced branch closures on the number of credit cards is smaller and statistically insignificant for individuals with inherited credit histories, while it is almost twice as large and more significant (top right panel) when compared to the average estimates in Figure 4. A significant difference is also observed for credit limits (bottom left panel), where individuals with inherited credit histories show none or even positive effects in the first nine years, compared to those without inherited histories. The bottom right panel shows the effect on the probability of having a *Good* credit score. Similarly, the results indicate that the negative impact of credit shocks is concentrated among individuals without an inherited credit history, who are 6 percentage points less likely to have a good credit score in the long run and exhibit a statistically significant negative impact. In contrast, individuals with an inherited credit history appear unaffected by the shock.

These results underline the importance of an established credit history in mitigating the negative consequences of credit supply shocks, suggesting that the opportunity to build a credit history is crucial for young individuals' credit score development.

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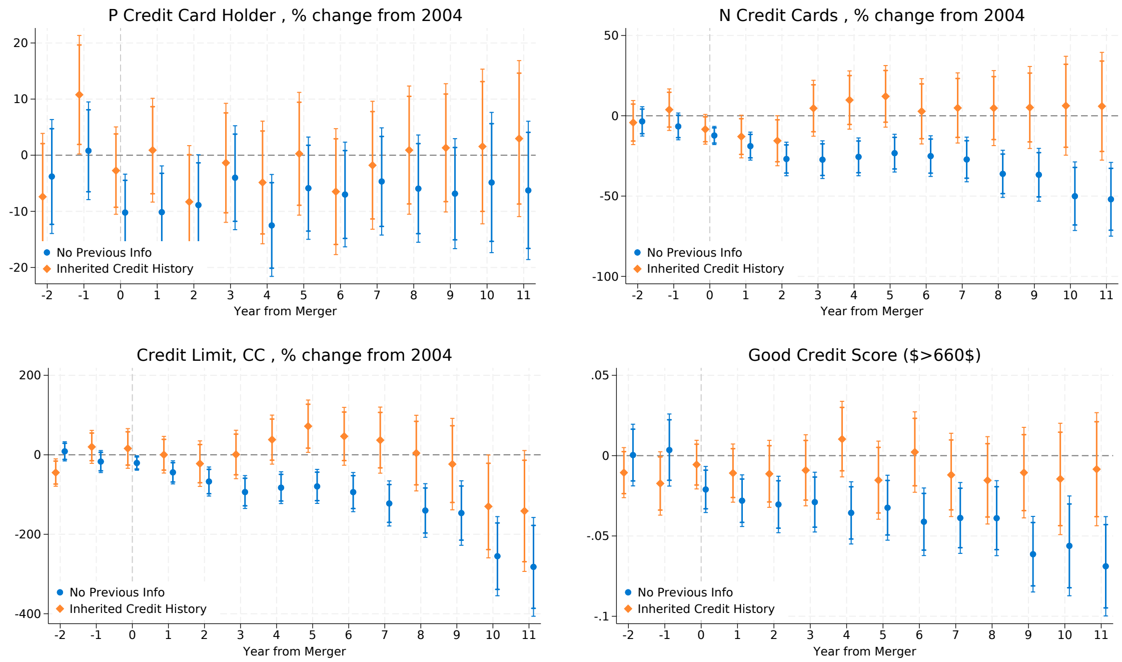
account of a customer with a longer credit history (potentially a parent). As documented in Bach et al. (2023), individuals who enter the market with an inherited credit history have significantly higher credit scores at entry and throughout their lifecycle.

Figure 10: The Effect of Credit Shock by Pre-existing Credit History



Each plot shows the estimated impact of a merger taking place at year $t = 0$ on different credit outcomes for individuals living in exposed ZIP codes. The effects are estimated separately for individuals with and without a pre-existing credit history. Estimates are based on model 2 and aggregated ATTs from mergers between July 2004 and June 2007, following Callaway and Sant'Anna (2021). Bars display 95% and 90% confidence intervals. The dependent variable is reported above each plot.

6 Conclusions

This paper investigates the long-term effects of shocks to early credit availability on subprime borrowers. We address a question that is fundamental to understanding the functioning of credit markets: what is the impact of initial conditions on long-term credit outcomes? Exploiting exogenous variation in credit supply due to merger-induced bank branch closures, we show that limited access to credit for young, low-credit-score individuals has large and persistent negative effects on future credit scores and borrowing. Our findings are consistent with the existence of a credit access trap, in which initial barriers to credit prevent individuals from building their credit scores and reinforce long-term exclusion from borrowing.

These findings carry important implications for policies promoting inclusion in consumer credit markets. If negative shocks to early credit access trigger a self-reinforcing cycle of exclusion, targeted interventions for young or disadvantaged individuals could yield substantial long-term returns—potentially even offsetting their costs. Reducing such early constraints may improve economic mobility and decrease inequality in borrowing opportunities over the lifecycle.

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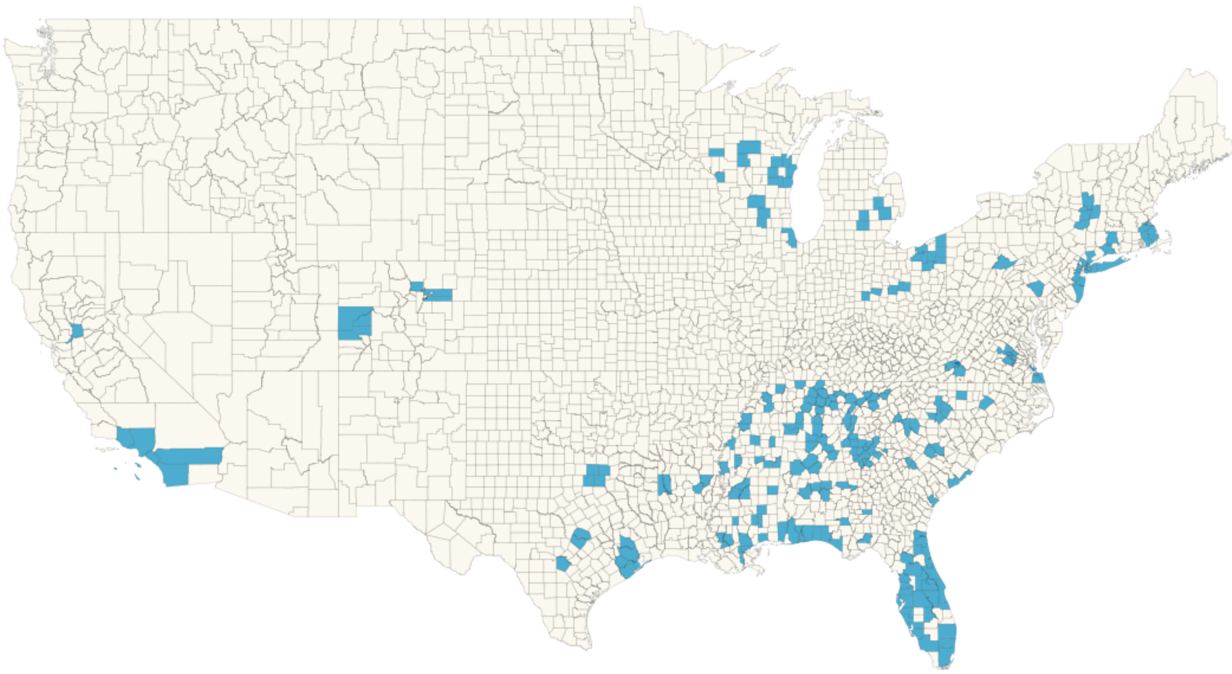
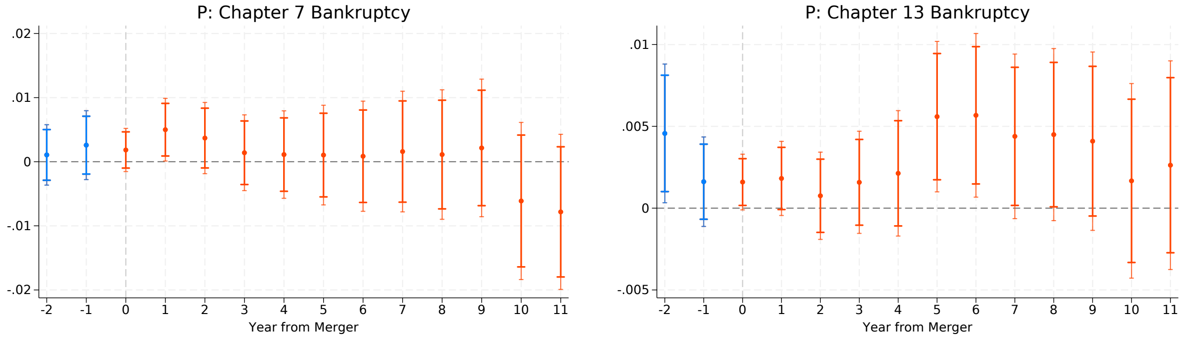
A Appendix

Figure 11: Overlapping Branches: Geographic Distribution



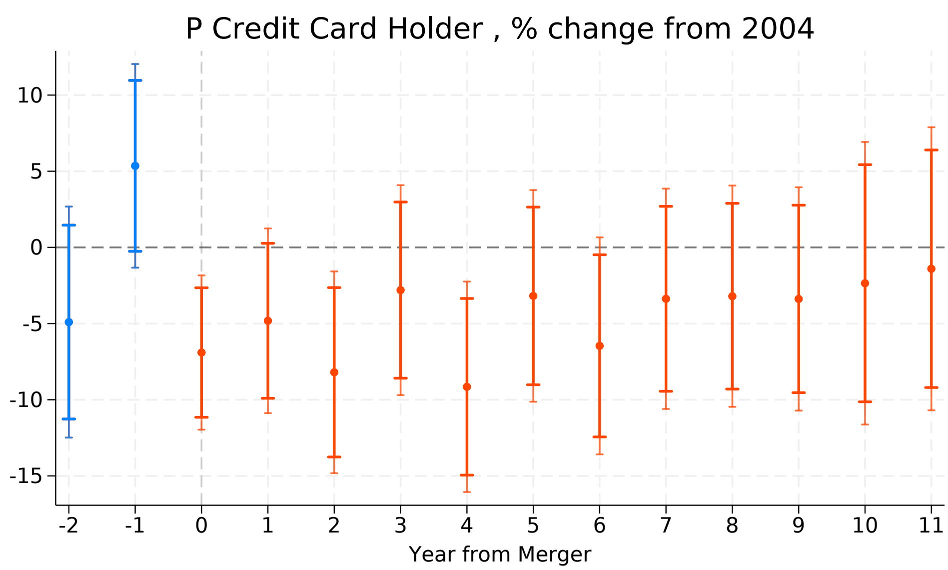
The map shows U.S. counties with overlapping branches from large banks involved in mergers between July 2004 and June 2007. Shaded areas indicate counties containing at least one ZIP code with branches from both the acquiring and acquired banks before the merger. Source: FDIC and authors' calculations.

Figure 12: Bankruptcies



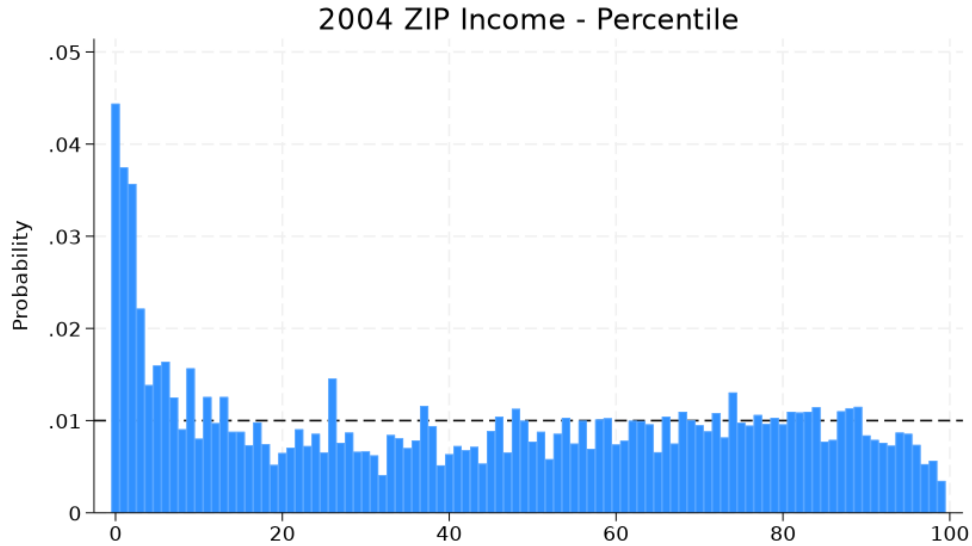
Each plot shows the estimated impact of a merger taking place at year $t = 0$ on different measures of bankruptcies for individuals living in exposed ZIP codes. Estimates are based on model 2 and aggregated ATTs from mergers between July 2004 and June 2007, following Callaway and Sant'Anna (2021). Bars display 95% and 90% confidence intervals. The dependent variable is reported above each plot.

Figure 13: Credit Card Holders



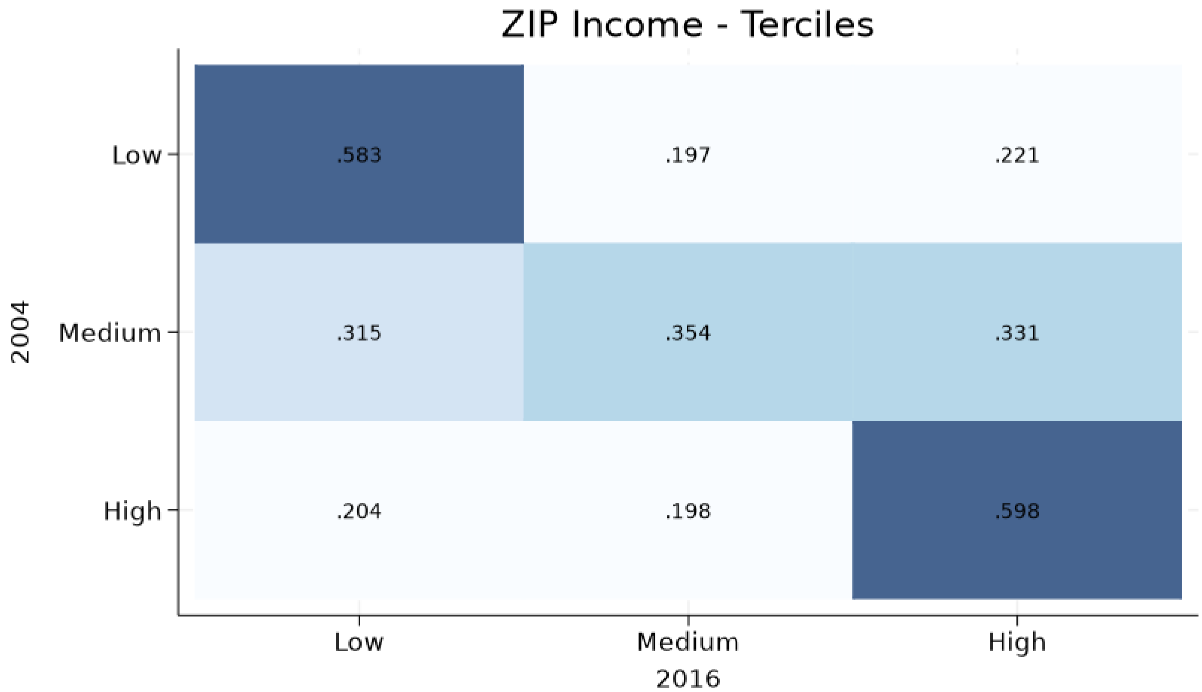
The plot shows the estimated impact of a merger taking place at year $t = 0$ on different measures of credit card holding for individuals living in exposed ZIP codes, expressed as a % change in the probability of holding a credit card in 2004, computed separately for treated and control units. Estimates are based on model 2 and aggregated ATTs from mergers between July 2004 and June 2007, following Callaway and Sant'Anna (2021). Bars display 95% and 90% confidence intervals.

Figure 14: Initial Distribution of Individuals by ZIP Code Income



The figure shows the distribution of individuals in the main sample across ZIP codes by percentile of the 2004 average ZIP code income distribution. Average ZIP code income is computed using publicly available IRS data.

Figure 15: Mobility Rates by ZIP Code Income



The figure shows the probability that individuals living in ZIP codes within each income tercile in 2004 (rows) reside in ZIP codes within each income tercile in 2010 and 2016 (columns). Darker shading indicates higher transition probabilities. The numbers in each cell report row-normalized probabilities of residing in a given 2016 income tercile, conditional on the 2004 tercile of residence.

Figure 16: Migration Rates, by Initial Income of ZIP Code of Residence

The figure shows the probability of living in a ZIP code different from that in 2004 (empty bars), as well as the probability of living in a different ZIP code in the bottom (orange) or top (blue) income tercile. ZIP codes are classified into percentiles of the year-specific income distribution, based on IRS data. The left graph refers to the population of individuals living in a ZIP code in the bottom tercile of the income distribution in 2004, while the right graph refers to individuals living in a ZIP code in the top tercile of the income distribution in 2004.

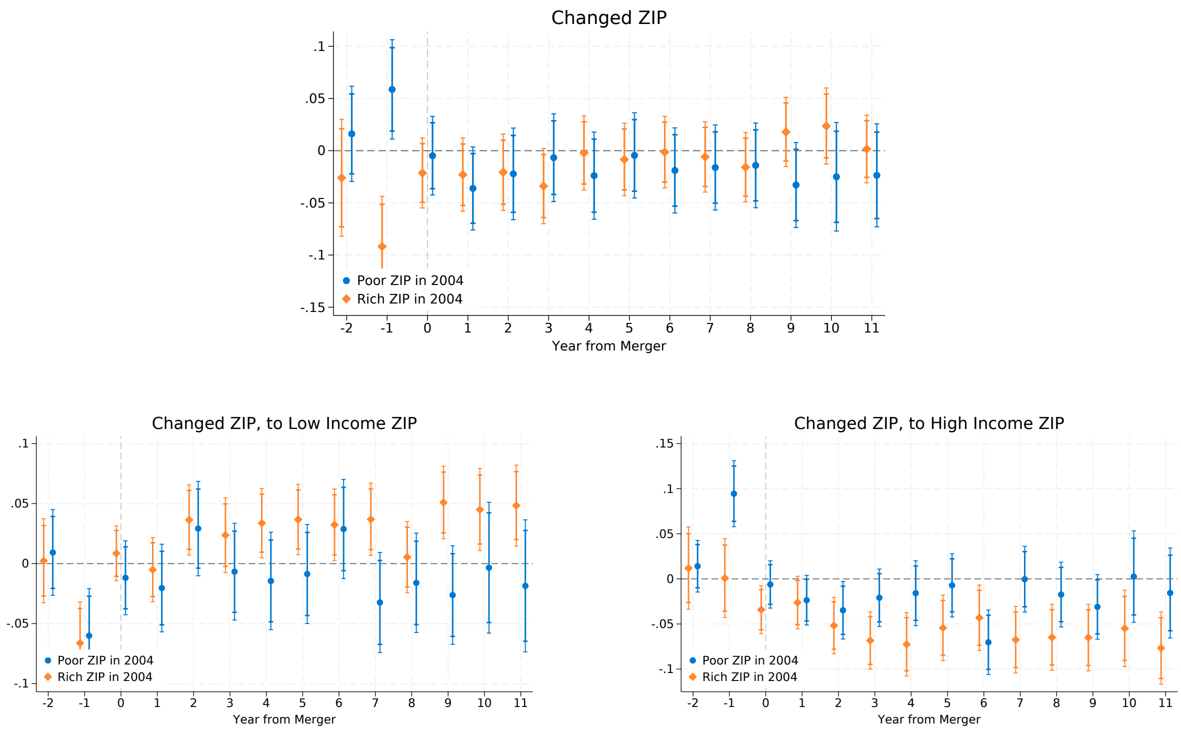


Figure 17: The Effect of Credit Shock on Geographic Mobility

The plot shows the estimated impact of a merger taking place at year $t = 0$ on different measures of geographic mobility for individuals living in exposed ZIP codes. The effects are estimated separately for individuals living in different ZIP codes at the top or bottom income tercile in 2004. Estimates are based on model 2 and aggregated ATTs from mergers between July 2004 and June 2007, following Callaway and Sant'Anna (2021). Bars display 95% and 90% confidence intervals. The dependent variable is reported above each plot.

Table 5: Summary Statistics of Credit Outcomes (2004–2016)

	2004		**2010**	**2016**
Exposed to Merger	*NO*	*YES*	*Both*	*Both*
C: Mortgages				
Mortgage: Access	0.01	0.01	0.10	0.19
	(0.11)	(0.12)	(0.30)	(0.40)
N of Open Mortgages	0.01	0.02	0.08	0.15
	(0.14)	(0.13)	(0.31)	(0.39)
Balance, MTA	1,455.89	1,386.24	10,882.38	25,067.00
	(16,052.70)	(12,616.86)	(46,546.15)	(76,880.83)
Original Loan, MTA	1,487.92	1,408.33	11,167.92	26,641.43
	(16,295.35)	(12,803.87)	(47,645.04)	(80,885.98)
Original Loan, MTA - Conditional on Access	7,779.47	6,672.21	57,368.40	136854.21
	(36,601.86)	(27,244.51)	(94,931.93)	(136117.41)
Amount 30d+ Delinquent, MTA	1.71	2.22	5.66	9.74
	(60.58)	(50.02)	(117.82)	(300.80)
P 60d+ Delinquent, MTA	0.99	0.98	0.92	0.86
	(0.12)	(0.12)	(0.26)	(0.35)
N Inquiries, MTA	0.17	0.15	0.40	0.50
	(0.59)	(0.53)	(0.92)	(0.98)
D: Auto Loans				
Auto Loan: Access	0.21	0.21	0.50	0.61
	(0.41)	(0.41)	(0.50)	(0.49)
N of Auto Loans (ever)	0.28	0.28	0.90	1.41
	(0.67)	(0.62)	(1.24)	(1.76)
Balance, AUT	1,712.11	1,571.03	3,533.68	7,236.53
	(5,034.61)	(4,503.10)	(7,619.13)	(12,747.98)
Original Loan, AUT	2,204.92	2,037.59	5,281.55	9,746.71
	(6,692.08)	(5,578.77)	(10,531.97)	(16,079.44)
Original Loan, AUA - Conditional on Access	3,006.16	2,831.27	8,481.34	16,038.97
	(7,086.04)	(6,489.36)	(12,352.58)	(18,015.13)
P 60d+ Delinquent, AUT	0.85	0.85	0.70	0.57
	(0.35)	(0.35)	(0.46)	(0.50)
N Inquiries, AUT	0.33	0.18	0.33	0.62
	(0.82)	(0.55)	(0.75)	(1.05)
Observations	18953	3937	22890	22890

Table 6: Summary Statistics by History at Entry (1)

	2004		2010		2016	
Inherited History	*YES*	*NO*	*YES*	*NO*	*YES*	*NO*
A: Customer Information						
Age	20.36	20.99	26.36	26.99	32.36	32.99
	(1.25)	(1.08)	(1.25)	(1.08)	(1.25)	(1.08)
Credit Score	526.95	525.21	546.90	561.52	576.73	591.85
	(47.02)	(49.28)	(81.70)	(78.58)	(92.76)	(90.61)
Good Credit Score (> 660)	0.00	0.00	0.11	0.12	0.19	0.23
	(0.00)	(0.00)	(0.32)	(0.33)	(0.40)	(0.42)
N Open Credit Lines	0.53	2.72	1.83	2.80	3.52	5.04
	(0.90)	(2.93)	(2.82)	(3.40)	(4.59)	(5.46)
Balance	1,541.32	8,756.20	15,217.81	24,959.30	36,218.89	53,361.54
	(8,631.44)	(24,487.52)	(45,007.18)	(59,079.18)	(78,987.23)	(98,627.12)
N Credit Lines 30d+ Delinquent	0.04	0.17	0.04	0.06	0.06	0.08
	(0.20)	(0.49)	(0.25)	(0.31)	(0.32)	(0.38)
N Inquiries	2.58	4.28	2.64	3.09	2.72	3.14
	(2.23)	(2.88)	(2.28)	(2.52)	(2.51)	(2.68)
B: Credit Cards						
P Credit Card Holder	0.23	0.52	0.29	0.41	0.48	0.63
	(0.42)	(0.50)	(0.45)	(0.49)	(0.50)	(0.48)
N Credit Cards	0.34	1.75	0.92	1.48	1.86	2.92
	(0.72)	(2.42)	(1.89)	(2.39)	(3.03)	(3.87)
Credit Limit, CC	179.85	1,685.81	1,842.59	2,688.14	5,826.57	9,233.09
	(690.04)	(4,337.14)	(5,611.36)	(6,867.78)	(13,083.47)	(17,610.33)
Utilization %, CC	92.87	81.64	53.40	59.67	52.15	53.77
	(46.90)	(52.03)	(44.50)	(41.19)	(38.21)	(37.57)
P Credit Card 90d+ Delinquent	0.12	0.52	0.16	0.22	0.13	0.17
	(0.33)	(0.50)	(0.37)	(0.42)	(0.34)	(0.38)
N Inquiries, CC	1.02	1.71	0.98	1.13	1.12	1.37
	(1.26)	(1.68)	(1.28)	(1.37)	(1.51)	(1.65)
Observations	12600	10290	12600	10290	12600	10290

Table 7: Summary Statistics by History at Entry (2)

	2004		2010		2016	
Inherited History	*YES*	*NO*	*YES*	*NO*	*YES*	*NO*
C: Mortgages						
Mortgage: Access	0.00	0.02	0.07	0.13	0.16	0.24
	(0.06)	(0.15)	(0.26)	(0.34)	(0.37)	(0.43)
N of Open Mortgages	0.00	0.03	0.06	0.10	0.13	0.17
	(0.07)	(0.19)	(0.27)	(0.34)	(0.37)	(0.42)
Balance, MTA	416.48	2,701.99	8,346.11	13,988.02	20,791.42	30,302.40
	(7,809.41)	(21,400.87)	(40,429.43)	(52,921.43)	(68,529.70)	(85,722.88)
Original Loan, MTA	419.97	2,765.16	8,529.86	14,398.19	22,054.89	32,257.60
	(7,859.73)	(21,752.16)	(41,082.73)	(54,445.34)	(72,078.58)	(90,197.33)
Original Loan, MTA - Conditional on Access	2,648.47	11,575.89	53,791.93	60,275.56	139084.90	135040.98
	(19,591.67)	(43,351.70)	(90,621.51)	(98,218.91)	(128420.54)	(142067.12)
Amount 30d+ Delinquent, MTA	0.56	3.32	4.41	7.19	9.90	9.54
	(22.75)	(84.13)	(112.50)	(124.00)	(377.60)	(163.39)
P 60d+ Delinquent, MTA	1.00	0.97	0.94	0.91	0.88	0.83
	(0.07)	(0.16)	(0.24)	(0.29)	(0.33)	(0.37)
N Inquiries, MTA	0.11	0.24	0.32	0.49	0.45	0.56
	(0.42)	(0.72)	(0.80)	(1.03)	(0.93)	(1.04)
D: Auto Loans						
Auto Loan: Access	0.09	0.36	0.41	0.61	0.55	0.68
	(0.28)	(0.48)	(0.49)	(0.49)	(0.50)	(0.46)
N of Auto Loans (ever)	0.09	0.51	0.65	1.21	1.18	1.70
	(0.31)	(0.87)	(0.99)	(1.43)	(1.58)	(1.91)
Balance, AUT	705.31	2,890.95	2,830.93	4,394.20	6,079.57	8,653.21
	(3,033.22)	(6,368.41)	(6,765.29)	(8,470.59)	(11,573.23)	(13,923.59)
Original Loan, AUT	827.37	3,827.69	4,257.32	6,535.71	8,144.48	11,708.62
	(3,491.68)	(8,632.53)	(9,405.05)	(11,644.19)	(14,591.89)	(17,534.21)
Original Loan, AUA - Conditional on Access	1,184.05	4,726.49	7,601.92	9,340.02	14,933.12	17,118.74
	(4,167.93)	(8,565.25)	(11,550.41)	(13,032.17)	(17,001.07)	(18,891.84)
P 60d+ Delinquent, AUT	0.93	0.76	0.75	0.64	0.61	0.51
	(0.25)	(0.43)	(0.43)	(0.48)	(0.49)	(0.50)
Observations	12600	10290	12600	10290	12600	10290

Table 8: Summary Statistics for Older Cohorts (23-30 in 2004) (1)

	2004	2010	2016
A: Customer Information			
Age	26.35	32.35	38.35
	(2.59)	(2.59)	(2.59)
Credit Score	624.98	635.14	656.43
	(99.97)	(110.17)	(108.99)
Good Credit Score ($>$ 660)	0.39	0.43	0.50
	(0.49)	(0.49)	(0.50)
N Open Credit Lines	4.95	4.75	5.56
	(4.75)	(4.45)	(5.12)
Balance	37,085.25	88,717.33	115507.30
	(79,435.75)	(157236.56)	(197591.56)
N Credit Lines 30d+ Delinquent	0.06	0.06	0.05
	(0.29)	(0.31)	(0.31)
N Inquiries	3.15	2.43	2.28
	(2.65)	(2.27)	(2.29)
B: Credit Cards			
P Credit Card Holder	0.68	0.65	0.73
	(0.47)	(0.48)	(0.44)
N Credit Cards	3.50	3.11	3.62
	(3.90)	(3.52)	(3.98)
Credit Limit, CC	9,749.79	14,076.90	21,530.15
	(15,688.04)	(21,429.41)	(29,955.81)
Utilization %, CC	38.70	36.46	36.87
	(40.10)	(37.18)	(34.16)
P Credit Card 90d+ Delinquent	0.20	0.17	0.11
	(0.40)	(0.37)	(0.31)
N Inquiries, CC	1.12	0.80	0.95
	(1.38)	(1.17)	(1.36)
Observations	144975	144975	144975

Table 9: Summary Statistics for Older Cohorts (23-30 in 2004) (31)

	2004	2010	2016
C: Mortgages			
Mortgage: Access	0.16	0.42	0.51
	(0.37)	(0.49)	(0.50)
N of Open Mortgages	0.21	0.44	0.46
	(0.53)	(0.75)	(0.72)
Balance, MTA	24,883.71	71,853.84	91,967.31
	(73,460.14)	(148522.23)	(186993.30)
Original Loan, MTA	26,306.92	76,386.14	102173.11
	(212571.09)	(157036.12)	(205408.59)
Original Loan, MTA - Conditional on Access	51,085.60	148334.77	198410.67
	(294079.29)	(192913.90)	(250679.07)
Amount 30d+ Delinquent, MTA	5.13	22.30	15.67
	(108.35)	(410.21)	(475.49)
P 60d+ Delinquent, MTA	0.82	0.66	0.63
	(0.38)	(0.47)	(0.48)
N Inquiries, MTA	0.59	0.63	0.54
	(1.16)	(1.12)	(1.00)
D: Auto Loans			
Auto Loan: Access	0.52	0.68	0.70
	(0.50)	(0.47)	(0.46)
N of Auto Loans (ever)	1.01	1.68	1.91
	(1.36)	(1.85)	(2.12)
Balance, AUT	4,946.77	5,577.10	8,418.31
	(9,241.31)	(10,558.57)	(14,922.79)
Original Loan, AUT	7,011.24	9,007.14	12,313.54
	(12,117.46)	(15,650.01)	(19,796.90)
Original Loan, AUA - Conditional on Access	9,037.52	12,685.69	17,518.70
	(13,270.89)	(17,328.88)	(21,596.31)
P 60d+ Delinquent, AUT	0.58	0.46	0.41
	(0.49)	(0.50)	(0.49)
Observations	144975	144975	144975